RANSAC

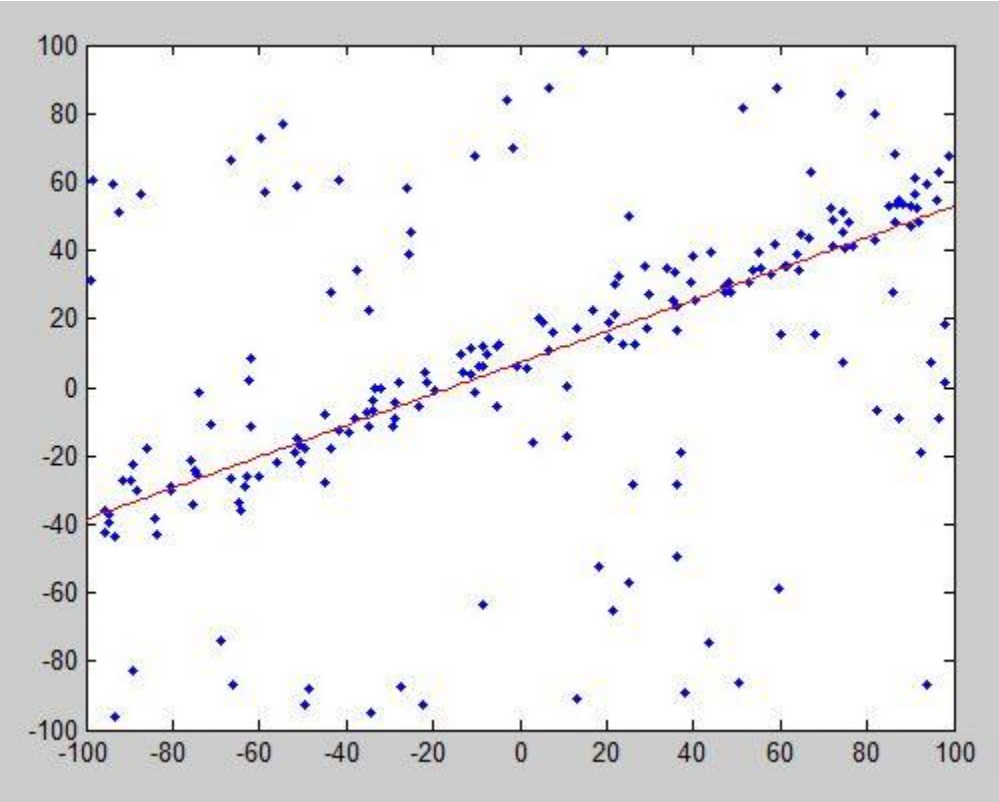
RANSAC

- Random sample consensus
- It is an iterative method for estimating a mathematical model from a data set that contains outliers.
- The RANSAC algorithm works by identifying the outliers in a data set
- It estimates the desired model using data that does not contain outliers.

Steps

- Randomly selecting a subset of the data set
- Fitting a model to the selected subset
- Determining the number of outliers
- Repeating steps 1-3 for a prescribed number of iterations

Data points shown in blue, with the line of form y = mx+c estimated using RANSAC indicated in red



- In computer vision, RANSAC is used as a robust approach to

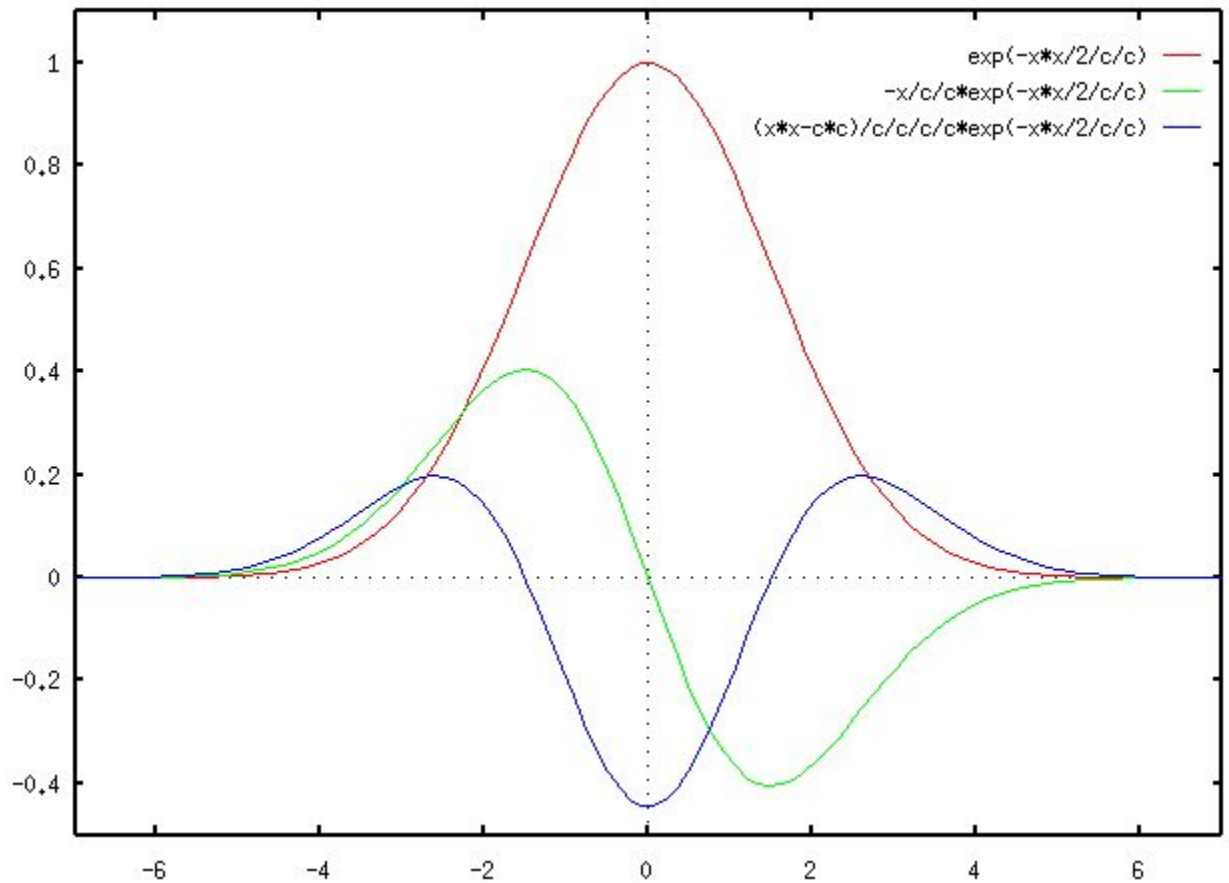
1. It estimates the fundamental matrix in stereo vision

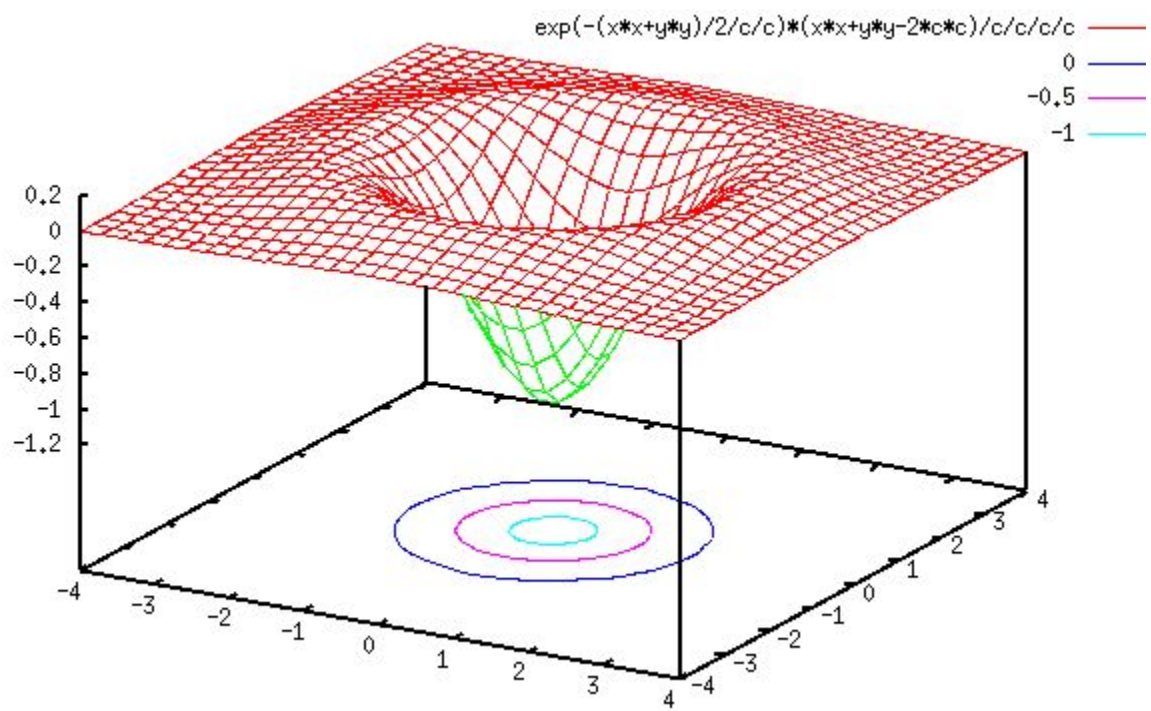
2. It finds the commonality between two sets of points for feature-based object detection, and

3. It registers sequential video frames for video stabilization.

Difference of Gaussian (DoG)

LoG


exp(-x*x/2/c/c)
-x/c/c*exp(-x*x/2/c/c)
(x*x-c*c)/c/c/c/c*exp(-x*x/2/c/c)
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-6
-4
-2
0
2
4
6


exp(-(x*x+y*y)/2/c/c)*(x*x+y*y-2*c*c)/c/c/c/c
0
-0.5
-1
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.2
-4
-3
-2
-1
0
1
2
3
4

DoG

- Similar to LoG, the image is smoothed by convolution with Gaussian kernel of certain sigma value σ_1

$$G_{\sigma_1}(x,y) = \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left(-\frac{x^2+y^2}{2\sigma_1^2}\right)$$

$$g_1(x,y) = G_{\sigma_1}(x,y) * f(x,y)$$

- Now the image is smoothed by convolution with Gaussian kernel of certain sigma value σ_2

$$G_{\sigma_2}(x,y) = \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left(-\frac{x^2+y^2}{2\sigma_2^2}\right)$$

$$g_2(x,y) = G_{\sigma_2}(x,y) * f(x,y)$$

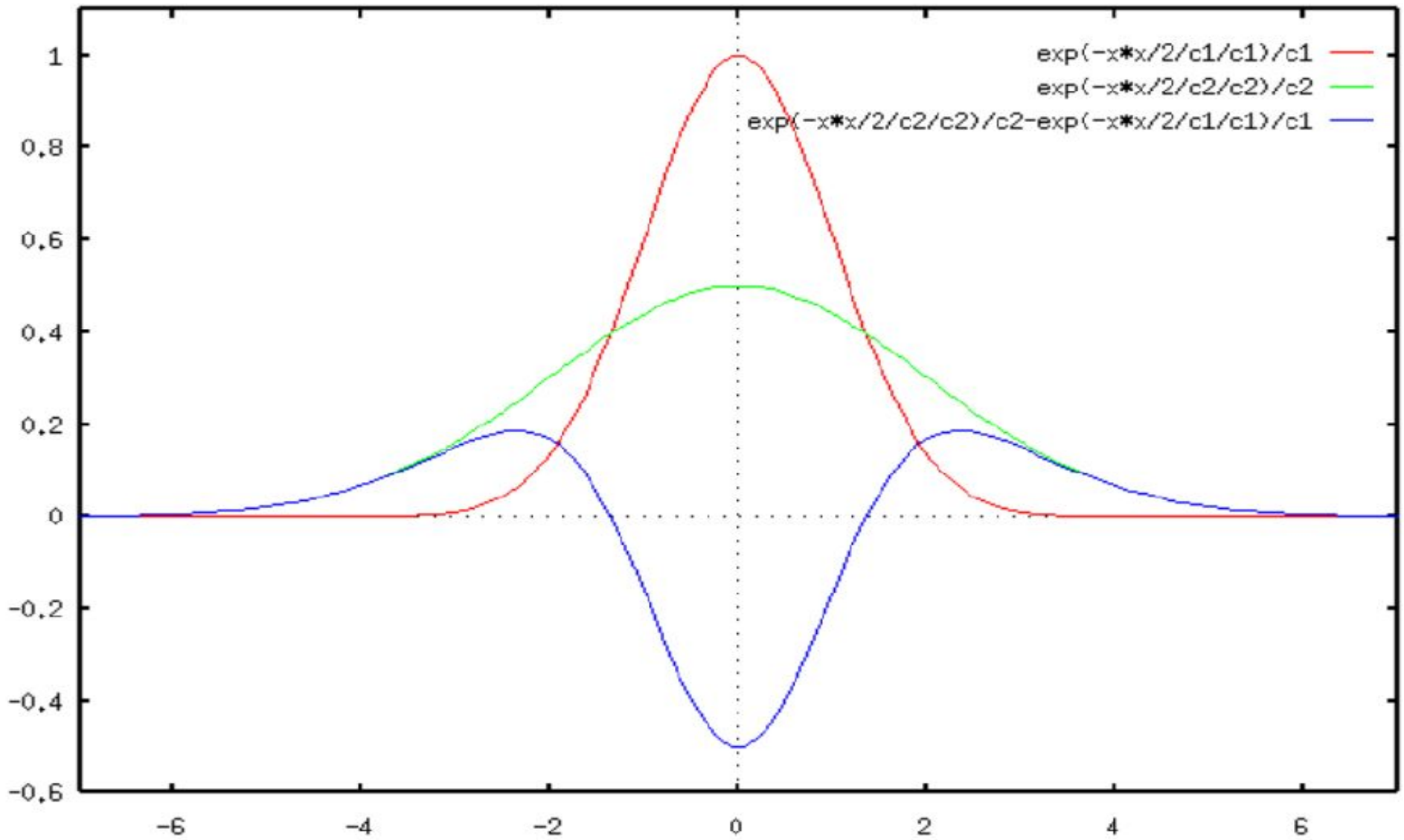
We can show that the difference of these two Gaussian smoothed images, called *difference of Gaussian (DoG)*, can be used to detect edges in the image.

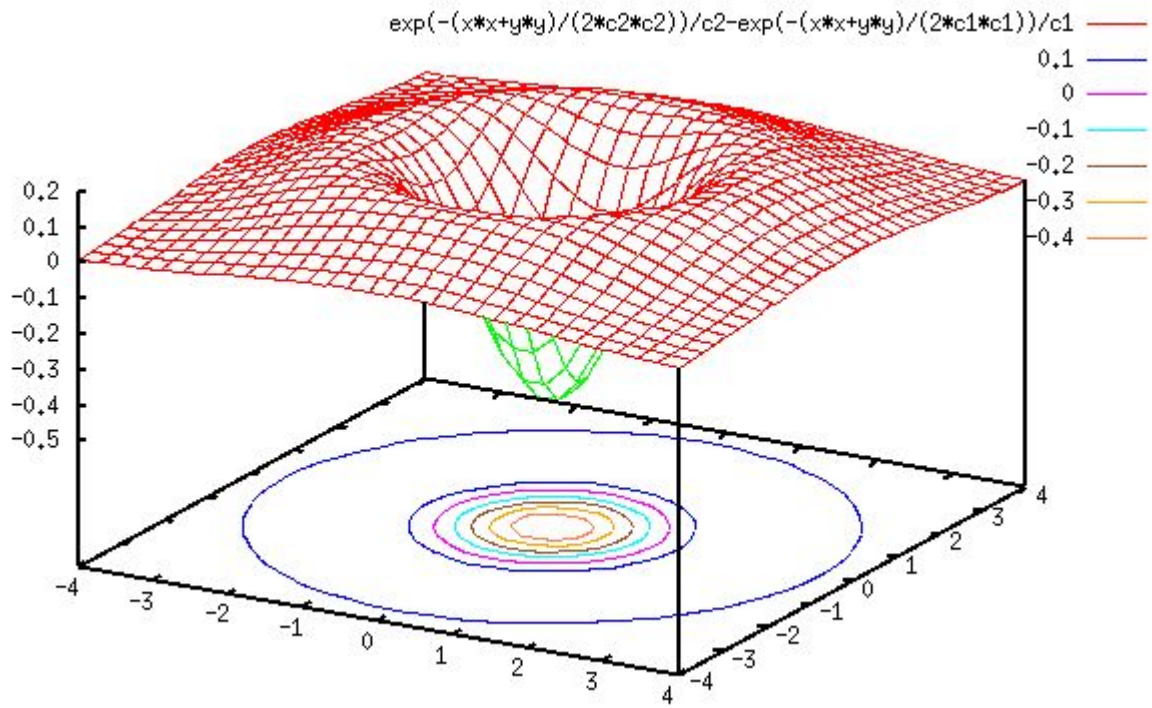
$$g_1(x,y) - g_2(x,y) = G_{\sigma_1} * f(x,y) - G_{\sigma_2} * f(x,y) = (G_{\sigma_1} - G_{\sigma_2}) * f(x,y) = DoG * f(x,y)$$

The DoG as an operator or convolution kernel is defined as

$$DoG \triangleq G_{\sigma_1} - G_{\sigma_2} = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{\sigma_1} e^{-(x^2+y^2)/2\sigma_1^2} - \frac{1}{\sigma_2} e^{-(x^2+y^2)/2\sigma_2^2} \right)$$

Both 1-D and 2-D functions of $G_{\sigma_1}(x, y)$ and $G_{\sigma_2}(x, y)$ and their difference are shown below:




exp(-(x*x+y*y)/(2*c2*c2))/c2-exp(-(x*x+y*y)/(2*c1*c1))/c1
0.1
0
-0.1
-0.2
-0.3
-0.4
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
-4
-3
-2
-1
0
1
2
3
4

- Comparing this plot with the previous one, we see that the DoG curve is very similar to the LoG curve.
- Also, similar to the case of LoG, the edges in the image can be obtained by these steps:
  - Applying DoG to the image
  - Detection of zero-crossings in the image
  - Threshold the zero-crossings to keep only those strong ones (large difference between the positive maximum and the negative minimum)
- The last step is needed to suppress the weak zero-crossings most likely caused by noise.

Corner Detection

- Image segmentation – dividing image in to regions
- Done on the similarity principle or on the basis of discontinuities
- Corner – important feature
- Landmark points
- Formed at boundary when two regions meet
- Boundary curvature is high
- Exhibits large variations in all directions
- Edges – variations in only one direction

- Known as interest points
- Human visual system quickly identifies the corner points
- Applications: live tracking, measurement systems, stereo imaging

Moravec Corner detector

- Traditional
- Calculates edge strength in all directions
- Minimum is selected
- Compared with the threshold
- Non maximal suppression is applied
- Extract set of corner points

Limitations

- Not rotationally invariant
- Sensitive to small variations

Harris corner detector

- Popular corner detection algorithm
- Uses autocorrelation function – find score with respect to direction
- Small square window is placed around each pixel
- Window is shifted in all 8 directions
- Auto correlation is performed
- No variation – region is a patch
- Large variation – marked as interest point

- Consider (x,y) as point
- Shifting of window (dx,dy)
- Corner score – c(x,y)
- C(x,y)=Image function – shifted image
- Shifted image can be approximated by taylor expansion